

# CWC 2.0 ESM — Business Flows

Traced from real code (backend/src/routes, controllers, services + backend/prisma/schema.prisma + prisma/seed.ts). All URL patterns mounted under /api/v1. Scope: **service desk / request platform (IT / HR / Finance / ESM-travel)** — CRM and credit excluded.

**Global gates:** all routes require authenticate; many use `authorizeResource(loadRequestScopeFromParam('id'), ...)` for row-level access (P02-09) plus `requestAccess.service.ts` policy evaluation. Admin routes use `requirePermission('admin:access')` (read) / `admin:settings` (write).

## 1. Service desks & catalog hierarchy

**Model hierarchy** (schema.prisma): `ServiceDesk (:424) → ServiceCategory (:454) → RequestType (:476)`. A desk belongs to a `Department (:342)` and has `autoAssignTeam`, `assignmentStrategy` (`ROUND_ROBIN`), `lastAssignedIndex (:434-438)`. `RequestType` is the catalog item: `requiresApproval`, `slaHours`, `JSON formConfig (+formConfigVersion)`, `workflowTypeId` (links a `WorkflowType` state machine), `lifecycleStatus` (catalog governance), `classification`, `ownerId`.

**Routes** (`routes/serviceDesk.routes.ts`): public reads at :22-50 (list desks/categories/request-types); admin CRUD gated by `admin:settings` at :103-172; deactivation-impact preview at :82-96.

**Service** (`services/serviceDesk.service.ts`): `getAllServiceDesks (:6)`, `getRequestTypes (:241)`, filters `lifecycleStatus: 'PUBLISHED', P5-01`, `reorderCategories` (transaction), deactivation-impact counts.

**Seeded desks** (`prisma/seed.ts`):

- **IT Support (IT) — 5 categories (:1031-1042):** Get IT Help (24h), Email Management (24h), Report System Problem (24h), Software Installation (48h), Request New Hardware (72h).
- **Group HR (HR) — 4 categories (:1162-1214):** Question for HR, New Hiring Request (48h), New Employee Onboarding, Offboard an Employee.
- **Group Finance (FINANCE) — 4 categories (:1292-1347):** Purchase Requisition, Inter-Company Chargeback, Submit Budget Proposal, Expense Claims (disabled). SLA 72h each.
- **Executive Services (ESM) — 1 category:** Travel Request (:1442-1463, 168h).

## 2. IT workflow

**Files:** `routes/it-workflow.routes.ts`, `controllers/it-workflow.controller.ts` (931 lines).

**Procurement chain** (via `transitionRequest`, guarded in `transitionGuards.ts` + `workflowTransitions.ts`):

`SUBMITTED → PROCUREMENT_IN_PROGRESS → HARDWARE_ORDERED → HARDWARE_RECEIVED → SOFTWARE_PROVISIONED → RESOLVED`, plus an executive chain `ACKNOWLEDGED_IT → PENDING_CEO_APPROVAL_IT → PENDING_CTO_APPROVAL_IT → PENDING_INVOICE_IT → PENDING_CFO_APPROVAL_IT → PAYMENT_PROCESSING_IT → PAYMENT_DONE_IT → PENDING_DELIVERY_IT → RESOLVED`.

- `markProcurement (:52)` — `SUBMITTED→PROCUREMENT_IN_PROGRESS`, stores order/vendor.
- `markHardwareOrdered (:128)` — `orderNumber/tracking → ITHardwareRequest`.
- `markHardwareReceived (:180)` — sets `procurementStatus: 'RECEIVED'` and **auto-creates an Asset** when `registerAsAsset && assetTag` (dedup on tag/serial, :210-238), inferring category via `inferCategoryFromName (:10-21)`.

- markSoftwareProvisioned (:264), markFulfilled (:98 → RESOLVED).
- Exec approvals: acknowledgeRequest (:311), ceoDecision (:410), ctoDecision (:534), routeToCfoApproval (:621, upload invoices), cfoDecision (:730), markPaymentDone (:819, branches hardware→PROCUREMENT vs software→PENDING\_DELIVERY\_IT), completeDelivery (:896).

**Model:** ITHardwareRequest (schema :1421) — hardwareName/model, estimatedPrice, vendor, manager approval flags, procurementStatus/order/tracking/serial/assetTag/assetId.

### 3. HR workflow

**Recruitment journey** (routes: approval, interview, screening, loa, onboarding, offboarding).

**Status chain** (workflowTransitions.ts :22-40 + :96-114):

SUBMITTED → PENDING\_CEO\_APPROVAL → CEO\_APPROVED → PENDING\_GROUP\_DCEO\_APPROVAL → GROUP\_DCEO\_APPROVED → JOB\_POSTED → PENDING\_MANAGER\_REVIEW → MANAGER\_APPROVED → INTERVIEW\_SCHEDULED → INTERVIEW\_FEEDBACK\_PENDING → HR\_SCREENING → LOA\_PENDING\_APPROVAL → LOA\_APPROVED → LOA\_ISSUED → LOA\_ACCEPTED → COMPLETED → ONBOARDING\_SUBMITTED → ... → ONBOARDING\_COMPLETED.

- **Job posting / CEO / Group DCEO** (approval.controller.ts): routeToCEO (:49), ceoDecision (:160, →CEO\_APPROVED→PENDING\_GROUP\_DCEO\_APPROVAL or skips if CEO is GROUP\_DCEO), markJobPosted (:356), routeToManager (:433), managerDecision (:569), entityDecision (:774), routeToGroupDceoHr (:922), groupDceoDecisionHr (:1024).
- **Interview** (interview.controller.ts): scheduleInterview (:10), updateInterviewSchedule, submitInterviewFeedback (:156, PROCEED/REJECT → INTERVIEW\_FEEDBACK\_PENDING). Models InterviewSchedule, InterviewFeedback.
- **Screening** (screening.controller.ts): startHRScreening (:10 → HR\_SCREENING), updateScreeningStatus (:111 → LOA\_PENDING\_APPROVAL). Model HRScreening.
- **LOA** (loa.controller.ts): upload → route-for-approval → manager-approve → mark-issued → upload-signed → mark-accepted (→LOA\_ACCEPTED → COMPLETED). Model LetterOfAcceptance.
- **Onboarding** (onboarding.controller.ts + services/onboarding.service.ts): createOnboardingFromHiring (:11) auto-derives hire from LOA/resume (start date +7 days); phases map to statuses (:199-223): ONBOARDING\_PENDING\_HR\_APPROVAL → PRE\_ARRIVAL\_SETUP → READY\_FOR\_DAY\_1 → DAY\_1\_ORIENTATION → WEEK\_1\_INTEGRATION → MONTH\_1/2/3\_MILESTONE → ONBOARDING\_COMPLETED. Models OnboardingRequest, OnboardingTask, OnboardingTaskTemplate.
- **Offboarding** (offboarding.controller.ts): →OFFBOARDING\_SUBMITTED → NOTICE\_PERIOD → KNOWLEDGE\_TRANSFER → FINAL\_WEEK → EXIT\_PROCEDURES → OFFBOARDING\_COMPLETED (:174-202); resignation-letter upload; IT revocation + HR flags. Models OffboardingRequest, OffboardingTask.

**Candidate:** Candidate (schema :1629) + CandidateResume (:1647), one-to-one with a hiring Request.

### 4. Finance workflow

**Files:** finance-workflow.routes.ts, chargeback-workflow.routes.ts; controllers finance-workflow.controller.ts (1310 lines), chargeback-workflow.controller.ts (434 lines).

**Purchase requisition** (finance-workflow.controller.ts): acknowledge (:152 → FINANCE\_ACKNOWLEDGED), routeToCfo (:187) / setFinalizedAmountAndRouteCfo (:261 →

PENDING\_CFO\_APPROVAL\_FIN), cfoDecision (:375; approve→CFO\_APPROVED\_FIN then branches FINANCE\_IN\_PROGRESS or PENDING\_GROUP\_DCEO\_APPROVAL :439-475), groupDceoDecision (:812 → GROUP\_DCEO\_APPROVED → PAYMENT\_PROCESSING\_FIN :855), markPaymentComplete (:912 → AWAITING\_PAYMENT\_CONFIRMATION), closeTicket (:950 → TICKET\_CLOSED\_FIN). Reassign approvers (:551, :696).

**Expense reimbursement** (same controller, routes :43-47): managerApproveExpense (:1031 → MANAGER\_APPROVED\_FIN) → financeHeadApproveExpense (:1154 → FINANCE\_HEAD\_APPROVED) → markExpensePaymentComplete (:1264 → PAYMENT\_COMPLETED → REIMBURSEMENT\_CLOSED). Models FinanceExpenseReimbursement (:1481) + ExpenseLineItem (:1513).

**Chargeback** (chargeback-workflow.controller.ts): submitChargeback (:23), fromEntityDecision (:101 → PENDING\_TO\_ENTITY\_APPROVAL / FROM\_ENTITY\_REJECTED), toEntityDecision (:219), markConfirmed (:348), completeChargeback (:393 → CHARGEBACK\_COMPLETED). Chain: PENDING\_FROM\_ENTITY\_APPROVAL → PENDING\_TO\_ENTITY\_APPROVAL → CHARGEBACK\_FINANCE\_REVIEW → AWAITING\_CHARGEBACK\_CONFIRMATION → CHARGEBACK\_COMPLETED.

## 5. ESM Travel workflow

**Files:** routes/esm-workflow.routes.ts (:17-23), controllers/esm-workflow.controller.ts (1063 lines).

**Approval chain** (workflowTransitions.ts :22-27): SUBMITTED → PENDING\_CEO\_APPROVAL → CEO\_APPROVED → PENDING\_GROUP\_DCEO\_APPROVAL → GROUP\_DCEO\_APPROVED → FINANCE\_ACKNOWLEDGED → PENDING\_CFO\_APPROVAL\_FIN → CFO\_APPROVED\_FIN → COMPLETED.

- submitForCeoApproval (:193) — routes to PENDING\_CEO\_APPROVAL, validates CEO/GROUP\_DCEO approver via executiveRole (:62-84, :171-174).
- ceoDecision (:442) — rejection→REJECTED; approval→CEO\_APPROVED; if the approver also holds GROUP\_DCEO it **auto-skips the DCEO stage** (:517-555), else →PENDING\_GROUP\_DCEO\_APPROVAL (:591).
- groupDceoDecision (:636 → GROUP\_DCEO\_APPROVED → FINANCE\_ACKNOWLEDGED :707), financeAcknowledge (:755 → PENDING\_CFO\_APPROVAL\_FIN), cfoDecisionTravel (:821 → CFO\_APPROVED\_FIN → COMPLETED :903, or reject→REJECTED).
- **Group DCEO threshold** configurable via systemSettingService.getEsmDceoThreshold (default 50000).

## 6. Approvals, delegation, policy

**Models** (Schema): ApprovalPolicy (:728) → ApprovalPolicyStep (:751) → ApprovalPolicyVersion (:782, frozen definition, DRAFT/PUBLISHED/RETIRED) → ApprovalInstance (:808) → ApprovalInstanceStep (:831). Approver types: ROLE/DEPARTMENT/ENTITY/USER/TEAM/AUTO (:690-697). Steps have parallelGroup + timeoutAction (REMINDER/ESCALATE/REJECT). Legacy RequestApproval (:1542) + ApprovalDelegation (:1593) + ApprovalReminder (:1613).

**Runtime** (services/approvalRuntime.service.ts): publishPolicyVersion (:88, freezes steps), retirePolicyVersion (:144), startApprovalInstance (:181, fail-closed — only PUBLISHED versions start; resolves policy by priority/tenant :194-205; activates sequential groups :229+), decideApproval (routes through executeWorkflowCommand), delegateApprovalRuntime (rejects self/cyclic/out-of-tenant delegation + requester SoD).

**Policy service** (approvalPolicy.service.ts): CRUD (:45), resolvePolicy (:121, lowest priority wins), step approver resolution (:171-209).

**Delegation service** (approvalDelegation.service.ts): delegateApproval (:26) verifies PENDING status + current approver, reassigns approverId→delegate, logs ApprovalDelegation; plus reminders and timeout fallback/escalation.

## 7. SLA & escalation

**Models** (schema): SlaPolicyVersion (:569, response/resolution/OLA targets in minutes, timezone Asia/Kuala\_Lumpur), SlaClock (:591), SlaPauseLedger (:620), SlaTimerJob (:636), SlaEscalationEvent (:665), EscalationRule (:521, triggerHoursAfterBreach + notifyRoles, per request type).

**Per-type SLA hours** from RequestType.slaHours seed (IT 24-72h, HR 48h, Finance 72h, ESM Travel 168h). Legacy Request carries slaDueAt, slaPausedAt, slaPauseDurationMs (:1232-1234).

services/sla.service.ts:

- checkSlaBreaches (:7) — overdue non-paused requests not in terminal statuses → creates a SYSTEM "SLA BREACH" activity (dedup), notifies assigned agent or first admin.
- checkEscalations (:89) — per breached request, loads matching EscalationRules ordered by triggerHoursAfterBreach; fires when breachedAt + hours ≤ now; idempotent via slaEscalationEvent upsert + outboxEvent; notifies notifyRoles and adds them as ESCALATION\_RECIPIENT participants (:185-219).

services/sla-pause.service.ts: pauseSla (:13, sets slaPausedAt, idempotent), resumeSla (:57, accumulates pause ms + extends slaDueAt), isPauseStatus (:131, reads WorkflowStep.slaPause, Redis-cached 5min), shouldResumeOnTransition (:163), getEffectiveSlaDueAt (:182), checkStalePauses (:226, auto-resumes after 14 days).

## 8. Notifications, assets, KB, search

- **Notifications** (services/notification.service.ts, 553 lines): notify (:463, IN\_APP+EMAIL inline), notifyDurably (:504, BullMQ queue), notifyMultiple (:537), publishDomainEvent (:127, writes NotificationDomainEvent + NotificationDelivery in tx), deliverNotification (:313), deliverPendingNotifications (:444). Global email toggle cached 30s. Models: Notification, NotificationTemplate, NotificationDomainEvent, NotificationDelivery.
- **Assets** (routes/asset.routes.ts, controllers/asset.controller.ts): full CRUD + CSV/XLSX import (parse/commit, :104-123), assignment (assignAsset), list active assignments, export. Models Asset (:2403, with sourceRequestId linking IT hardware), AssetAssignment (:2445). Auto-provisioning in IT workflow markHardwareReceived.
- **KB** (routes/kb.routes.ts, controllers/kb.controller.ts): published-article reads + admin create/update/delete/publish; scoped by policy visibility (kb:manage). Model KnowledgeBaseArticle (:1813, per service desk, helpful/not-helpful counts, viewCount). **No separate kb.service.ts** — logic lives in the controller.
- **Search** (routes/search.routes.ts, controllers/search.controller.ts): globalSearch (:10) across Requests, KB Articles, Users — department-scoped via policyService.buildVisibleWhere (P02-11, :30-31); plus searchRequests, searchArticles, searchUsers.

## 9. Cross-flow notes for maintainers

- **Runtime transitions are DB/published-workflow driven**; the hardcoded `VALID_TRANSITIONS` map is a seed/fallback only (see 03-request-lifecycle.md).
- **Many workflow services are embedded in controllers** — there is **no** `it-workflow.service.ts`, `kb.service.ts`, `screening.service.ts`, `interview.service.ts`, or `loa.service.ts`; logic lives in the controllers (verified).
- `prisma/db-dump.json` **and** `dist/` **are build artifacts**, not sources of truth.
- **SLA/escalation, notification delivery, and asset provisioning** are cross-cutting; changes here affect every workflow.